

James Matthews

Astrophysicist and Royal Society URF

Department of Physics (Astrophysics)
University of Oxford
Oxford, OX1 3RH
☎ (+44)7933139071
james.matthews@physics.ox.ac.uk
jhmatthews.github.io

Academic Career

- 2022–present **Royal Society University Research Fellow**, *University of Oxford*
Project: “Extreme Cosmic Accelerators: High-energy particles and feedback from black hole jets and winds”
- 2019–2022 **Herchel Smith Fellow**, *Institute of Astronomy, University of Cambridge*
Independent Postdoctoral Fellowship, “The origin and physics of the highest energy particles in nature”
- 2016–2019 **Postdoctoral Researcher**, *University of Oxford*
Project title: “The origin of ultra-high energy cosmic rays”, PIs: Profs. A. R. Bell and K. M. Blundell
- 2012–2016 **PhD Physics**, *University of Southampton*
Thesis: “Disc Winds Matter: Modelling Accretion and Outflow On All Scales”, Supervisor: Prof. C. Knigge
- 2012 **Research Year Abroad**, *Harvard-Smithsonian CfA*
Thesis: “Searching For Nearby Planets During Predicted Microlensing Events”, Supervisor: Dr. R. Di Stefano
- 2008–2012 **MPhys Astrophysics**, *University of Southampton*, first-class honours

Research Interests & Goals

- Building models for particle acceleration and transport in magnetised plasmas.
- Studying the physics and observational signatures of accretion discs and their associated outflows.
- Understanding the connection between outflows and galaxy evolution.
- Developing state-of-the-art radiative transfer and MHD methods to complement observations.
- Uncovering the origin of the highest energy cosmic rays (CRs).
- Testing exotic particle physics (e.g. axions) with astrophysical probes.

Programming Skills

- Advanced Python, C, Fortran, git/github, travis-CI, L^AT_EX, OpenMPI Parallelisation, VisIt.
- Familiar IDL, Topcat, Bash, MCMC methods.
- Astro (magneto)hydrodynamics, Monte Carlo radiative transfer, CLOUDY, CRPROPA, PLUTO, TARDIS, XSPEC.
- Lead SIROCCO, MSYNCHRO, ALPRO, CR-VAR-JET, CR-REVERB.
- Developer

Awards, Courses & Successful Proposals

- 2024–present **Green Templeton College, Research Associate**
- 2025 **UKRI HPC Facilities**, *co-PI*, 10 Million CPU Hours
Successful UKRI proposal on Isambard 3 (with DPhil student Emma Elley)
- 2023 **Royal Society Leadership Effectiveness course**
Successful completion of three-day course on effective leadership for academics, run by Imperial Business School
- 2022 **Royal Society URF Enhanced Research Expenses**, *PI*, £400K
Funding for 2x 4-year PhD Studentships and a two-year PDRA (with potential for extension)
- 2022 **Royal Society URF**, *PI*, approx. £700K
5 + 3 year fellowship awarded by the Royal Society to scientific leaders of the future, including overheads, indirect costs, travel, salary at 0.8 FTE.
- 2019 **HST Proposal**, *co-I*, *UV spectroscopy of a super-Eddington BHXT (abbr.)*, PI: Castro-Segura
- 2019 **Herchel Smith Fellowship**, approx. £150K
Independent fellowship at Cambridge for three years. Value estimate based on combined salary and expenses.

- 2018 **LBS Proposal, co-I**, *Laboratory model of particle acceleration in Supernova shocks*, PI: Chen
- 2017 **HST Proposal, co-I**, *Wide band spectra of nova-like variables*, PI: Long
- 2016 **Springer Thesis Prize for Outstanding PhD Research**, *University of Southampton*
- 2015 **SALT Proposal, PI**, *The spectra of nova-like variables*
- 2013 **RAS Grant**, *Visit to Columbia University*
- 2008-2012 **Academic Scholarship**, *Top 5 students in physics, University of Southampton*

Service and Collaborations

- 2023-present WEAVE-LOFAR collaboration member
- 2022-present ThunderKAT collaboration member
- 2023-present Summer Internship Co-ordinator, Oxford Astrophysics
- 2023-present DPhil Graduate Careers co-organiser, Oxford Astrophysics
- 2016-present Referee: AAS Journals, MNRAS, PASA, A&A, Nature Astronomy
- 2023-2024 DPhil Admissions Panel, Oxford Astrophysics
- 2023 Local Organising Committee, ThunderKAT Meeting, Oxford
- 2023 DFG (German Research Foundation), expert reviewer
- 2023,2024 Science Technology Facilities Council grants, expert reviewer
- 2022-2023 Swiss National Science Foundation, expert reviewer
- 2022-2023 Liverpool Telescope, expert reviewer
- 2020-2022 IoA Forum on Racial Equality in Astronomy, co-founder
- 2020-2022 IoA Equality & Diversity Committee Member
- 2020-2022 IoA Student Career Panel
- 2021 EUCapt White Paper: Convener (Cosmic Accelerators)
- 2021 EUCapt White Paper: Contributor (Travelling Messengers)
- 2020 IoA Lunchtime Seminar Organiser
- 2019 Oxford women in physics career advice panel
- 2018 Invited reviewer for “100 years of jets” review anthology.
- 2018 Astrophysics summer project admissions and co-ordination
- 2017-2019 SPI-MAX seminar organiser
- 2017-2019 Galaxies coffee organiser
- 2017 Local Organising Committee, Plasma Astrophysics Conference, Oxford
- 2015 Local Organising Committee, TORUS 2015, Winchester

Teaching, Mentoring & Public Engagement

- 2023-present Graduate Lecture: Particle Acceleration
- 2018-present Oxford Astrophysics Outreach for the Homeless, Organiser and founder
Leadership of the first public engagement program to bring astronomy and physics into Oxford homeless shelters (O'Hanlon House). As a result, formed ongoing departmental partnership with Homeless Oxfordshire.
- 2024 Oxford Astrophysics Flagship Open Doors event at Green Templeton College
- 2023 Into The Cosmos, Oxford Astrophysics
- 2020-2022 IoA Student-Postdoc Liason Officer
Duties included organising the weekly journal club, practice talks and acting as a conduit between the student and postdoc bodies (50+ students and 70+ postdocs).
- 2020-2021 Low Participation Neighbourhood Summer School Speaker (Physics), Cambridge University
Delivery of lecture course on special relativity to high-achieving 16 year olds from from areas of low progression to higher education with a view to raising aspirations and increasing progression to highly selective universities.
- 2021 EAS 2021: Mentor
- 2021 Public Talk, Cambridge University Astronomy: **Through the Amber Spyglass**
- 2020 Public Talk, Cambridge Astronomy on Tap: **Spooky Jets from Black Holes**

- 2020 Meet an Astronomer, Cambridge IoA lockdown engagement project
- 2018-2019 Undergraduate Tutor, Astrophysics C1
Tutored the 4th-year undergraduate astrophysics course at Oxford, with tutorials rated 5/5 by all respondents.
- 2016 AstroAirport Demonstrator, University of Southampton
- 2015, 2016 SETI Cipher Challenge Helper, University of Southampton
- 2012-2016 10+ school visits as Astrodome Demonstrator, University of Southampton
- 2012-2016 Teaching Demonstrator, University of Southampton
Courses: Energy and Matter, Quantum Physics of Matter, Python for Scientific Computing, Introduction to Scientific Computing

Student Supervision

- 2024-present Clara Lilje, Oxford (DPhil co-supervisor)
- 2024-present Charlotte Jackson, Oxford (DPhil co-supervisor)
- 2023-present Emma Elley, Oxford (DPhil supervisor)
- 2022-present Katie Savard, Oxford (DPhil co-supervisor)
- 2025-present Cainã de Oliveira, University of Sao Paolo (Visiting Student sponsor)
- 2024-present Alessandra Ambrifi, IAC (Visiting Student sponsor)
- 2020-2024 Julia Sisk Reynes, IoA Cambridge (PhD co-supervisor)
- 2020-2021 Amy Rankine, IoA Cambridge (PhD informal co-supervisor role)
- 2021-2022 Part III/Masters supervisor: Henry Whitehead (2020-21), Jago Strong-Wright (2021-22)
- 2020-2023 Masters co-supervisions: Alex Granovskiy (2020-21), Katie Savard (2021-22), Nicky Bacon (2023-24)
- 2017-2023 Summer project supervisions: Ziyang Li (2017), Andrew Sellek (2018), Daniel Baig (2023)

Press and Media

- 2023 Scientific American, Interviewed/Quoted, *“The Second Most Powerful Cosmic Ray in History”*
- 2023 New Scientist, Interviewed/Quoted, *“We are finally closing in on the origins of the OMG article”*
- 2023 Appearance on BBC Oxford Radio, *“Into The Cosmos”*
- 2022 NASA Press Release, *“Black Hole spins slower than its peers”*
Featured on Sky & Telescope, ABP Live, The Independent, Space.com+
- 2019 RAS Press Release, *“Repeating outflows of hot wind found close to black hole”*
Featured on CNN, Science Daily, Environmental News Network+
- 2018 Appearance on BBC Oxford Radio, *“How do Mobile Phones Work?”*

Talk Highlights

At least 60 talks (25+ invited) at international conferences and colloquia in Europe, Asia, USA, selection below

- 2025 **Invited Talk**, National Astronomy Meeting, Durham
- 2025 **Invited Talk**, Vasto Accretion Meeting, Vasto, Italy
- 2025 **Invited Colloquium**, University of Durham
- 2024 **Invited Talk**, EAS, Padova
- 2024 **Invited Colloquium**, University of Manchester
- 2024 **Invited Colloquium**, University College London
- 2023 **Invited Talk**, European Plasma Conference, Bordeaux
- 2023 **Invited Colloquium**, IIHE, Brussels
- 2023 **Contributed Talk**, AGN On The Beach, Tropea
- 2023 **Contributed Talk**, Microquasar 2023, Crete
- 2022 **Invited Review**, 23rd European Cosmic Ray Symposium, Nijmegen
- 2021 **Invited Colloquium**, University of Bristol (remote)

2021 **Invited Review**, *HIRSAP Summer School* (remote)
 2021 **Invited Talk**, *IoP UK CTA Meeting* (remote)
 2021 **Invited Review**, *EUCapt Symposium, CERN* (remote)
 2020 **Invited Colloquium**, *University of Amsterdam* (remote)
 2020 **Invited Review**, *EAS 2020, Leiden* (remote)
 2019 **Invited Review**, *A Centenary of Astrophysical Jets, Manchester*
 2019 **Invited Colloquium**, *University of Bath*
 2019 **Invited Colloquium**, *University of Glasgow*
 2018 **Invited Talk**, *Hillas Symposium, Heidelberg*
 2018 **Invited Talk**, *Particle Acceleration and Transport, Calabria*
 2018 **Invited Colloquium**, *Queen's University, Belfast*
 2018 **Contributed Talk**, *UHECR 2018, Paris*
 2018 **Invited Colloquium**, *University of Manchester*
 2017 **Invited Review**, *Broadband Astrophysical Processes, Southampton*
 2017 **Invited Colloquium**, *University of Southampton*
 2017 **Contributed Talk**, *AGN Winds on the Georgia Coast, Jekyll Island, GA*
 2015 **Contributed Talk**, *TORUS 2015, Winchester*
 2015 **Invited Talk**, *The Golden Age of CVs, Palermo*
 2015 **Contributed Talk**, *Black Hole Accretion and AGN feedback, Shanghai*

References

Prof. Christian Knigge, *University of Southampton*, C.Knigge@phys.soton.ac.uk
 Connection: PhD Supervisor

Prof. Rob Fender, *University of Oxford*, rob.fender@physics.ox.ac.uk
 Connection: Sub-department head/line manager and collaborator

Prof. Tony Bell FRS, *University of Oxford/Rutherford-Appleton Lab*, tony.bell@physics.ox.ac.uk
 Connection: Former PI at Oxford.

Prof. Christopher Reynolds, *University of Maryland*, csr12@ast.cam.ac.uk
 Connection: Collaborator at Cambridge.

Prof. Paul Hewett, *University of Cambridge*, phewett@ast.cam.ac.uk
 Connection: Collaborator at Cambridge.

Publications

Please see [ADS publications](#) for full list (would not fit on four-page CV format).

85 articles (15 first-author, 18 second author) published in journals such as *Monthly Notices of the Royal Astronomical Society*, *The Astrophysical Journal*, *Nature*, *Physical Review D*, *New Astronomy Reviews*.

H-index: 27, citations: 1916 (ADS) 2237 (Google Scholar).